PROG POE

Prog6212

St10481943

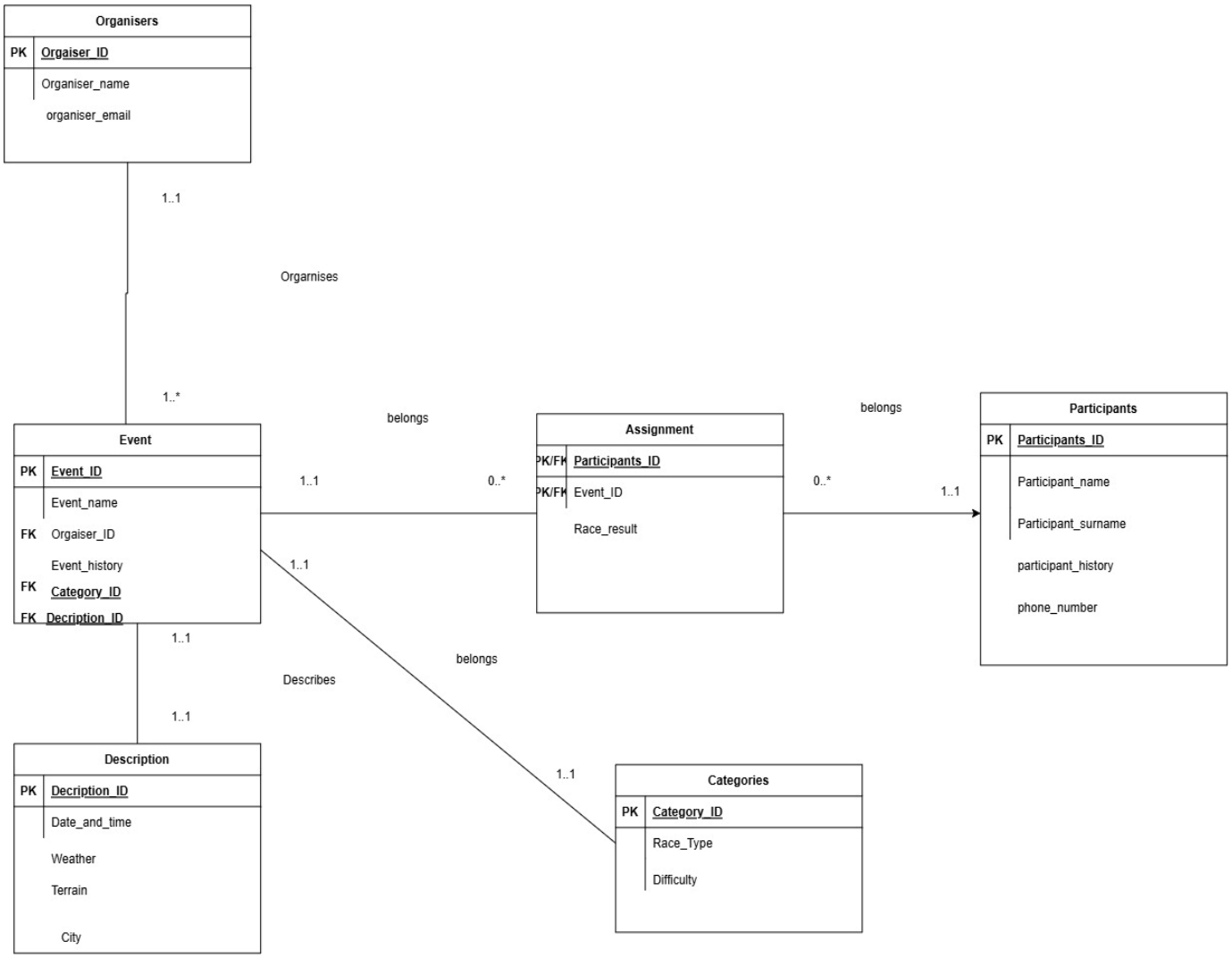
Mighty Mlahleki

Contents

Section A 1

Section B 1

Section A



Section B

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/auth/register`	Register a new user account.	None	`{ name, email, password }`	`201 Created` – User

					registered. 400 Bad Request
POST	/api/auth/login	Authenticate a user and return a token.	None	{ email, password }	200 OK – JWT token returned. 401 Unauthorized
GET	/api/profile	Retrieve logged-in user's profile.	Any (Logged In)	None	200 OK – User profile data
PUT	/api/profile	Update logged-in user's profile.	Any (Logged In)	{ name, phone, etc. }	200 OK – Profile updated
GET	/api/categories	Retrieve all event categories.	None	None	200 OK – List of categories
POST	/api/categories	Create a new category.	Admin	{ name, description }	201 Created – Category created
GET	/api/categories/{id}	Retrieve a specific category.	None	None	200 OK – Category details
PUT	/api/categories/{id}	Update a category.	Admin	{ name, description }	200 OK – Category updated
DELETE	/api/categories/{id}	Delete a category.	Admin	None	200 OK – Category deleted
GET	/api/events	Retrieve all events.	None	None	200 OK – List of events
POST	/api/events	Create a new event.	Admin	{ title, description, date, venue, categoryId }	201 Created – Event created
GET	/api/events/{id}	Retrieve a specific event.	None	None	200 OK – Event details
PUT	/api/events/{id}	Update an event.	Admin	{ title, description, date, venue, categoryId }	200 OK – Event updated
DELETE	/api/events/{id}	Delete an event.	Admin	None	200 OK – Event deleted
POST	/api/events/{id}/enroll	Enroll a user in an event.	Any (Logged In)	None	201 Created –

					Enrollment successful
GET	`/api/enrollments`	View logged-in user's enrollments.	Any (Logged In)	None	`200 OK` – Enrollment list
GET	`/api/enrollments/{id}`	View a specific enrollment record.	Any (Logged In)	None	`200 OK` – Enrollment details

Section C

-- creating database

create database RaceDayDB;

-- using the created database

use [RaceDayDB];

--creating tables for the database

create table organisers(organiser_ID int primary key identity (1,1),

organiser_name varchar(50) not null, organiser_email varchar(50)not null);

create table Description_Race(Description_ID int primary key identity(1000,1),

Date_and_Time Datetime not null, Weather varchar(50) not null, Terrain varchar(50)not null, City varchar(50)not null);

create table Participants(Participants_ID int primary key identity (1100,1),

participant_name varchar(50)not null,participant_surname varchar(50)not null, participant_history varchar(50)not null,

phone_number varchar(10)not null);

create table categories(Category_ID int primary key identity (2100,1), Race_Type varchar(50)not null,

Difficulty varchar(50)not null);

```
create table Race_Event(Event_ID int primary key identity (100,1), Event_name varchar(50)not null,

Event_history varchar(50) not null, organiser_ID int not null, Category_ID int not null,Description_ID int not null,

Constraint FK_Race_Event_organisers FOREIGN key(organiser_ID) REFERENCES organisers(organiser_ID),

FOREIGN key(Category_ID) REFERENCES categories(Category_ID),

FOREIGN key(Description_ID) REFERENCES Description_Race(Description_ID));


create table Assigment(Participant_ID int not null, Event_ID int not null, Race_Result varchar(50) not null,

constraint FK_History_Race_Event Foreign key(Event_ID) references Race_Event(Event_ID),

 Foreign key(Participant_ID) references Participants(Participants_ID));


-- inserting values into organisers

insert into [organisers]

values

( 'John Doe', 'johnD@events.co.za'),

( 'Sarah Williams', 'sarah@events.co.za'),

( 'David Jones', 'david@events.co.za');



--inserting values into Categories

insert into [categories]

values

('Marathon', 'Hard'),

('Trail Run', 'Medium'),

('Fun Run', 'Easy');



--inserting values into Descriptions
```

```
insert into [Description_Race]
values
('2026-10-10 08:00:00', 'Sunny', 'Road', 'Durban'),
('2026-11-15 07:30:00', 'Cloudy', 'Mountain', 'Johannesburg'),
('2026-12-05 09:00:00', 'Sunny', 'Road', 'Polokwane');


-- Events
insert into [Race_Event]
values
('Comrades marathon', 'Annual marathon', 1, 2100, 1000),
('Cradle Run', 'Annual trail running event', 2, 2101, 1001),
('Polokwane Fun Run', 'Community fun run', 3, 2102, 1002);


--inserting values into Participants
insert into [Participants]
values
( 'Thabo', 'Mokoena','Experienced runner', '0711111111' ),
( 'Lerato', 'Molefe','Trail runner', '0722222222' ),
( 'Peter', 'Smith','First-time participant', '0733333333'),
( 'Nomsa', 'Dlamini', 'Marathon runner', '0744444444'),
( 'James', 'Nkosi','Fun run participant', '0755555555');


-- Assigning participants to events
insert into [Assigment]
values
(1100, 100, '1st'),
(1101, 100, '2nd'),
(1102, 100, '3rd'),
```

(1101, 101, '1st'),

(1103, 101, '2nd'),

(1104, 102, '1st')

```
83
84    select * from [Assigment];
85    select * from [categories];
86    select * from [Race_Event];
87    select * from [Participants];
88    select * from [Description_Race];
89    select * from [organisers];
```

100 % No issues found

Results Messages

3	102	Polokwane Fun Run	Community fun run	3	2102	1002

	Participants_ID	participant_name	participant_surname	participant_history	phone_number
1	1100	Thabo	Mokoena	Experienced runner	0711111111
2	1101	Lerato	Molefe	Trail runner	0722222222
3	1102	Peter	Smith	First-time particip...	0733333333
4	1103	Nomsa	Dlamini	Marathon runner	0744444444
5	1104	James	Nkosi	Fun run participant	0755555555

	Description_ID	Date_and_Time	Weather	Terrain	City
1	1000	2026-10-10 08:00:00.000	Sunny	Road	Durban
2	1001	2026-11-15 07:30:00.000	Cloudy	Mountain	Johannesburg
3	1002	2026-12-05 09:00:00.000	Sunny	Road	Polokwane

	organiser_ID	organiser_name	organiser_email
1	1	John Doe	johnD@events.co.za
2	2	Sarah Williams	sarah@events.co.za
3	3	David Jones	david@events.co.za

Query executed successfully.

Ready

```
83
84    select * from [Assigment];
85    select * from [categories];
86    select * from [Race_Event];
87    select * from [Participants];
88    select * from [Description_Race];
89    select * from [organisers];
```

100 % No issues found

Results Messages

	Participant_ID	Event_ID	Race_Result
1	1100	100	1st
2	1101	100	2nd
3	1102	100	3rd
4	1101	101	1st
5	1103	101	2nd
6	1104	102	1st

	Category_ID	Race_Type	Difficulty
1	2100	Marathon	Hard
2	2101	Trail Run	Medium
3	2102	Fun Run	Easy

	Event_ID	Event_name	Event_history	organiser_ID	Category_ID	Description_ID
1	100	Comrades marathon	Annual marathon	1	2100	1000
2	101	Cradle Run	Annual trail running event	2	2101	1001
3	102	Polokwane Fun Run	Community fun run	3	2102	1002

Links

https://youtu.be/3rQAVOhyUI4

https://github.com/ST10481943/Prog6212-POE.git